

Research Article

Using PIT tags to infer bat reproductive status and parturition date: busy nights during lactation

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Abstract

Passive integrated transponders (PIT tags) can aid in the collection of important demographic data for species for which other methods, such as GPS technology, are not suitable. PIT tags can be particularly suitable to monitor small and cryptic species like bats and permit inference on their behavioral ecology. Literature for several species of bats states that females change their nightly activity patterns—going out and in of the roost only once per night during gestation compared to several times during lactation. Hence, we tested whether PIT tag detection patterns could be used to infer reproductive status and parturition date of female bats. From 2017 to 2021, we recorded detections of PIT-tagged little brown (*Myotis lucifugus*) and northern long-eared bats (*M. septentrionalis*) at the entrance of 2 maternity roosts in Québec and Newfoundland, Canada. We also used the maternity roost in Québec as a case study to further evaluate the potential of this method to link behavior and demography and investigated factors affecting parturition date. We were able to infer reproductive status for 63% to 97% of tagged individuals detected during both the gestation and the lactation periods, and parturition date for 61% to 95% of reproductive individuals, depending on the year and roost. Early spring arrival at the roost and warm spring mean temperature at night were associated with earlier parturition dates. Herein, we highlight that PIT tag systems may be useful to detect changes in activity patterns of female bats and infer individual reproductive parameters, which is on the long-term less stressful for bats and easier for researchers. We demonstrate that this approach is useful to investigate intrinsic and extrinsic factors of reproductive parameters, improving our understanding of bat population dynamics resulting in more informed population management decisions.

Key words: maternity roost, *Myotis lucifugus*, *Myotis septentrionalis*, parturition, passive integrated transponder, reproductive success, temperature, timing of arrival.

Uso de PIT tags para inferir el estado reproductivo de los murciélagos y la fecha de parto: Noches ajetreadas durante la lactancia

Resumen

Los transpondedores pasivos integrados (PIT tags) pueden ayudar en la recolecta de datos demográficos importantes de especies para las que otros métodos como la tecnología GPS no son adecuados. Los PIT tags pueden ser adecuados especialmente para el seguimiento de especies pequeñas y crípticas como los murciélagos y permiten inferir sobre su ecología comportamental. Según la literatura, las hembras de varias especies de murciélagos cambian sus patrones de actividad nocturna de una salida y regreso al refugio durante la gestación, a varias salidas y regresos durante la lactancia. Por lo tanto, probamos si los patrones de detección de PIT tags pueden utilizarse para inferir el estado reproductivo de los murciélagos hembra y las fechas de parto. Entre 2017 y 2021, registramos detecciones del murciélago pardo pequeño (*Myotis lucifugus*) y del murciélago de orejas largas del norte (*Myotis septentrionalis*) marcados con PIT tags, en la entrada de dos refugios de maternidad en Quebec y Terranova, Canadá. También utilizamos el refugio de maternidad de Quebec como estudio de caso y evaluar, con más detalle, el potencial de este método para vincular comportamiento y demografía, e investigamos los factores que afectan la fecha de parto. Inferimos el estado reproductivo de entre el 63 y el 94% de los individuos marcados detectados durante los períodos de gestación y lactancia, y la fecha de parto de entre el 61 y el 95% de los individuos reproductivos, dependiendo del año y del refugio. La llegada temprana a los refugios y la temperatura cálida promedio durante la noche en primavera se asociaron con fechas de parto más tempranas. Por lo tanto, resaltamos que los sistemas de marcaje PIT tag pueden ser útiles para detectar cambios en los patrones de actividad de los murciélagos hembra e inferir parámetros reproductivos individuales, lo cual es, a largo plazo, menos estresante para los murciélagos y más fácil para los investigadores. Demostramos que este enfoque es útil para investigar los factores intrínsecos y extrínsecos de los parámetros reproductivos, mejora nuestra comprensión de la dinámica de las poblaciones de murciélagos y se espera que pueda emplearse en la toma de decisiones de manejo de las poblaciones.

Palabras clave: éxito reproductivo, momento de llegada, *Myotis lucifugus*, *Myotis septentrionalis*, parto, refugio de maternidad, temperatura, transpondedor pasivo integrado.

Demographic studies investigating population growth rates, lifespan, reproductive histories, and social associations of wildlife require a capacity to identify individuals while limiting impacts on behavior and survival (Barclay and Bell 1988; Kunz and Weise 2009). Several new technologies emerged during the last few decades to track wild animals, yet few tools are suitable for small cryptic animals. Passive integrated transponders (PIT) are small microchips encased in inert glass. These tags can be inserted subcutaneously and emit a unique code when excited by an electromagnetic field. PIT tags are small enough to apply to small vertebrates and have become a popular means of permanently marking individuals (Fiedler 2009; Rigby et al. 2012; Omeyer et al. 2019). They have been used on many taxa to answer questions about animal physiology, behavior, social interactions, phenology, movement, and population dynamics (Gibbons and Andrews, 2004; Nomano et al. 2014; Soanes et al. 2015; Schorr and Siemers 2021).

PIT tags have been particularly useful for research on bats and have facilitated the acquisition of valuable information about bat population dynamics including phenology, apparent survival, site fidelity, relative abundance, and colony size (O'Shea et al. 2011; Santos-Moreno et al. 2021; Schorr and Siemers 2021). However, such studies have been limited in their inference of reproductive success. Indeed, other methods including telemetry, visual counts, or in-hand capture-mark-recapture are still required to evaluate these parameters (e.g. Zahn 1999; O'Shea et al. 2010; Franci et al. 2012). Finding ways of simplifying the collection of information relative to reproductive success would contribute to improved knowledge of bat ecology. For species having a high roost fidelity during the reproductive season, such as the Little Brown Bat (*Myotis lucifugus*) using buildings as maternity roosts (Lewis 1995; Norquay et al. 2013; Schorr et al. 2021), PIT tag detection patterns may be an efficient means to infer individual reproductive status and parturition date.

We know that reproductive females of several bat species have different nightly activity budgets while gestating and lactating (Swift 1980; Anthony et al. 1981; Maier 1992; Horn 1998; Henry 2001). Anthony et al. (1981) first documented changes in time budgets of reproductive little brown bats based on observations inside a maternity roost coupled with periodic captures. Using an infrared automatic counter installed at the roost to record the number of bats entering and leaving throughout the night, Maier (1992) also found a unimodal nightly activity pattern in pregnant pipistrelle bats (*Pipistrellus pipistrellus*), changing to a bimodal pattern in lactation, and returning to a unimodal pattern in postweaning. In another study, Henry (2001) used data from periodic captures, telemetry, and videos of little brown bats at a maternity roost to show that gestating females usually have 2 foraging bouts separated by 1 resting bout outside the roost. On the other hand, lactating females typically have 2 or 3 bouts away from the roost separated by 1 or 2 returns to the maternity roost, likely to feed their pup (Henry 2001). Even with an imperfect PIT tag detection caused by having only 1 antenna for multiple roost exits, the nightly time budget of 199 PIT-tagged little brown bats showed clear and repeatable patterns illustrating several bouts away from the roost per night during the lactation period (Horn 1998). Therefore, using PIT tag detections may be a means to reliably infer individual reproductive status and parturition date, and better understand underlying factors that explain changes in these traits without having to physically recapture individuals.

Adverse weather can influence the reproductive success of bats by reducing pregnancy rates, delaying parturition, and slowing juvenile growth (Racey and Swift 1981; Grindal et al. 1992;

Linton and Macdonald 2018). Insectivorous bats living in northern temperate environments are particularly susceptible to adverse weather due to their relatively small size and dependence on prey whose activity is influenced by weather (Anthony et al. 1981; Grindal et al. 1992; Lewis 1993). Torpor is used by insectivorous bats to minimize energetic costs during unfavorable environmental conditions. However, the use of torpor by females during spring and summer may slow fetal development, consequently delaying parturition, and reducing milk production (Racey 1973; Wilde et al. 1999). For healthy populations, the consequences of delayed spring on reproductive output are part of environmental stochasticity and likely have little impact on long-term demographics (Gaillard et al. 2000; but see Doak et al. 2005). However, for declining populations that face several threats, such as North American bats with white-nose syndrome (WNS; Cheng et al. 2021), it becomes important to understand these dynamics and it would be preferable to do so without further disturbing them via physical recaptures. Timing of parturition may also be related to other factors, such as the timing of arrival at the roost in spring, age, body condition, or genetics (Forslund and Larsson 1992; Holroyd 1993; Fulin et al. 2009). As found for some birds, females arriving earlier may be in better condition, allowing an earlier parturition date (Fulin et al. 2009).

The main goal of this study was to test the potential of data collected from a PIT tag monitoring system to infer individual reproductive status and parturition date. We predicted that if roosts of bat species known to change their activity patterns during gestation compared to lactation are efficiently monitored, PIT tag detections should enable inferences on reproductive status and parturition date. We tested these predictions at 1 maternity roost hosting little brown (*M. lucifugus*) and northern long-eared bats (*M. septentrionalis*) in Québec and with another roost hosting little brown bats in Newfoundland. To further evaluate the potential of this method at improving knowledge on bat population ecology, we chose the maternity roost in Quebec as a case study. We analyzed 5 years of data from PIT-tagged female bats and used inferred parturition for each individual to investigate whether variation in parturition dates could be explained by annual weather conditions and individual arrival date at the roost. We predicted a positive relationship between spring arrival date to maternity roosts and parturition dates. We also predicted later parturitions during wet, cold, and windy springs. Over the 5 years of monitoring, we expected to find individuals with earlier parturition dates to be consistently earlier across years due to intrinsic factors such as body size, body condition, or age.

Materials and methods

Study area

We PIT-tagged bats at 1 maternity roost in Québec and another in Newfoundland, Canada (Fig. 1). The Domaine Joly (DJ) roost in Québec was the attic of a historical building, converted into a restaurant and surrounded by a 19th-century garden park of 138 hectares on the south shore of the Saint Lawrence River (46.66677, -71.84898). The mean daily roost group size at the DJ during this post-WNS study was approximately 45 little brown and northern long-eared bats (this colony used to be over 1,000 individuals before WNS arrival in 2012). The roost in Newfoundland was in the Salmonier Nature Park (SNP) and included 11 bat boxes positioned within 1 km² (47.26270, -53.28488). The SNP bat boxes hosted a colony of >800 little brown bats, and there was no record of WNS prior to or during this study. The 2 sites had a cool

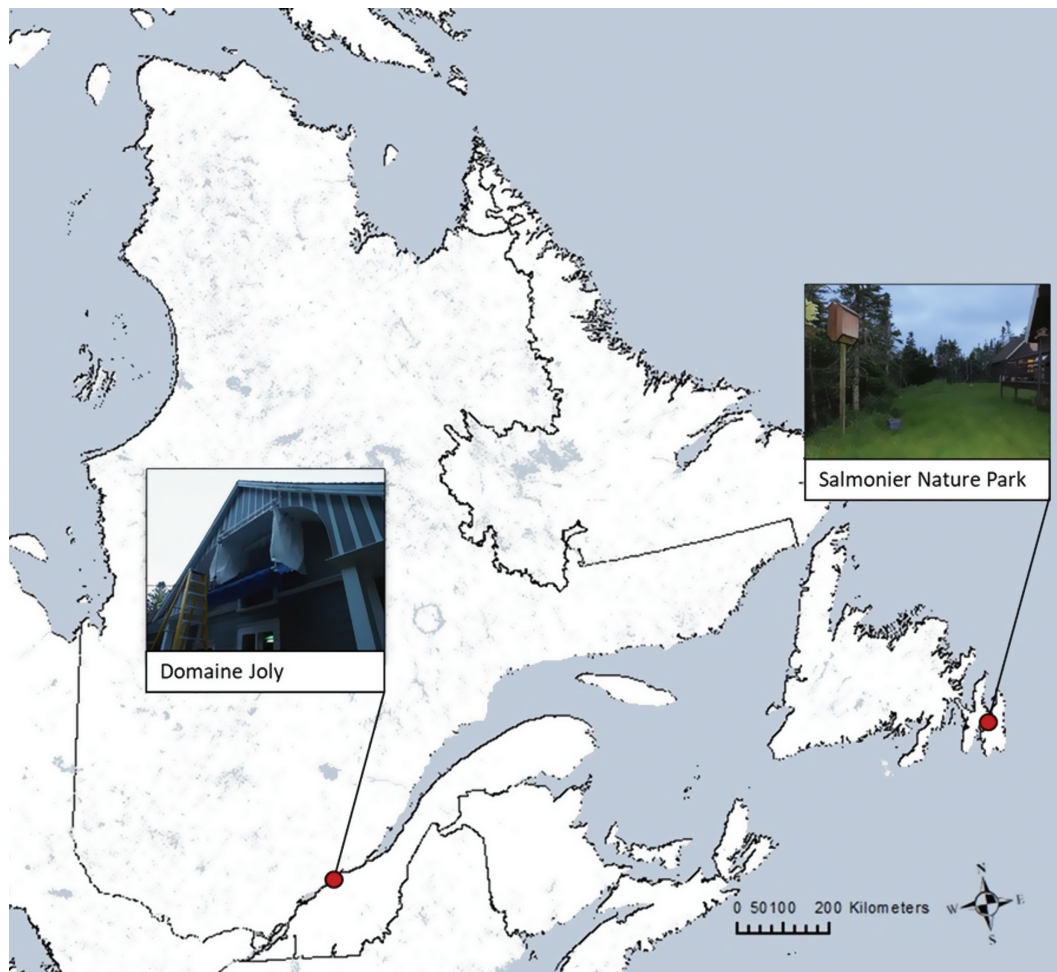


Fig. 1. Map showing the location of 2 maternity roosts where we PIT-tagged bats and recorded their detections, from 2017 to 2021 in Quebec and Newfoundland, Canada. Map derived from https://www12.statcan.gc.ca/census-recensement/2011/geo/bound-limit/files-fichiers/2016/lpr_000b16a_e.zip, modified with ArcGIS 10.4.1. software (www.arcgis.com).

temperate climate, with a mean temperature in June of 16 °C at the DJ and 14.5 °C at the SNP.

Bat PIT tagging

We captured and marked little brown (*M. lucifugus*) and northern long-eared bats (*M. septentrionalis*) after parturition in July, when young were believed to be born but not yet volant at the DJ roost in 2016, 2017, and 2018. We also captured bats before parturition in early June in 2017. At the SNP roost, we marked bats several times a year since 2012. We PIT-tagged a total of 53 bats (41 little brown bats and 8 northern long-eared bats) at the DJ roost and 1,425 little brown bats at the SNP roost. For each bat captured, we recorded the species (*M. lucifugus* or *M. septentrionalis*) depending on the tragus shape and the ear length, as well as body mass, sex, and reproductive condition by palpating the abdomen or examining the nipples (Racey 2009). We marked bats with GPT12 (12.5 mm) PIT tags (Biomark, Inc., Boise, Idaho) in Québec and with ID-100s (12.0 mm) PIT tags (Trovan, Ltd, The Netherlands) in Newfoundland. We inserted tags subcutaneously between the scapula, and sealed insertion sites with a biomedical glue (Vetbond Tissue Adhesive; 3 M Science, St. Paul, Minnesota).

To detect bats, we used antennas and reader systems powered by household AC power. We used a 0.8 m × 0.3 m window-style PIT tag antenna framing the only known entrance with an

HPR Plus reader (FS2001; Biomark, Inc.) at the DJ and a 0.40 m × 0.15 m window-style PIT tag antenna at all bat box entrances (Trovan Ltd.) at the SNP roost. We installed antennas in early spring and removed them in fall, usually before bats arrived and after they left the roost. Antennas and reader systems detected PIT tag signals throughout the study period while bats were active at the roost (generally from mid-May to mid-September). We constrained time allowed between 2 consecutive detections on the reader system to 30 s for a given individual. Based on previous knowledge, this interval between 2 detections of a given individual was the best compromise to avoid recording too many detections of a same individual resting near the antenna, without missing “independent” detections (Fontaine A, McGill University, Montreal, Quebec, Canada, personal communication, 1 April 2017).

Inference of reproductive status and parturition date

We sorted PIT tag detections by individuals. As bats sometimes wait beside the antenna and were recorded several times at the roost entrance—even with our 30-s interval setting—we kept only 1 detection per individual within each 15-min interval. We kept the last detection of the first sequence of consecutive detections (i.e. all detections within a 15-min interval from one another) in early evening as the time that the bat first left the

roost at night, and the first detection of the last sequence of consecutive detections when it entered the roost for the last time in the early morning. We cleaned detection sequences between the first exit and the last entrance, alternating with exits and entrances.

We inferred reproductive status of tagged females that stayed during the period of gestation and lactation (i.e. from at least June to July). We identified a female as reproductive if we observed an exit in the evening and a return in the morning per night without mid-evening return for 7 consecutive days preceding a change in detection patterns, followed by 2 or more emergence sequences per night without a single emergence sequence in the 7 following days. Days without detections were allowed, as long as days with detections within the 7 days before and after the change followed this pattern (Fig. 2a). We concluded that the individual did not give birth and for discussion, identify them as nonreproductive when only single emergence sequences were detected during the whole season (Fig. 2b). We assessed an individual as unknown reproductive status if it stopped being recorded before the end of the mean parturition date, if it did not fit our inference criteria, or if both periods contained too much variability or too many missing data, acknowledging that individuals may have given birth elsewhere or behave differently than expected. For females identified as reproductive, we inferred the parturition date as the date when there was a change in the activity pattern; i.e. at least 3 consecutive nights with a single emergence sequence followed by at least 3 consecutive nights with at least 2 emergence sequences (days without detections were not allowed to ensure precision in the parturition date; Fig. 2a).

Phenology and determinants of parturition date

At the DJ roost, we analyzed the phenology of reproduction including spring arrivals, inferred parturition dates, inferred decreased parental care onsets, and fall departures. As nightly

activity pattern was found to be unimodal during the postweaning period (Maier 1992), we inferred onset of the decrease in parental care as the date where the female transitioned from a multi-emergence sequence back to a single sequence. We assessed spring arrival and fall departure from the first and last detection of an individual at the roost, respectively, acknowledging that females could have arrived earlier or remained in a nearby alternative roost that was not monitored. Little brown and northern long-eared bats in our study shared the same roost and environmental conditions, and are known to have similar juvenile growth (i.e. age of full flight abilities; Krochmal and Sparks 2007). Therefore, after a visual inspection of the data suggesting no clear distinction between the 2 species in terms of reproductive status or parturition date, we pooled the 2 *Myotis* species together. To investigate the impact of environmental factors on reproduction, we calculated mean and minimal temperature (°C), total and mean rainfall (mm), rain duration (hour), and mean wind speed (km/h) at night using the hourly data from Deschambault meteorological station of Environment Canada, located 9 km from the DJ roost (https://climate.weather.gc.ca/historical_data/search_historic_data_e.html). Spring weather included data from 1 March to 30 June.

Statistical analysis

We calculated the proportion of bats for which we could assess reproductive status and parturition date for PIT-tagged females that were detected during both the gestation and the lactation periods at the DJ roost ($n_{\text{tot}} = 20$ to 27) from 2017 to 2021 and at the SNP roost in 2017 (a subsample of 60 bats were selected to validate the method). For the DJ roost, we used females for which we were able to infer of reproductive status and parturition date for further analyses. We evaluated whether variation in parturition date was influenced by spring weather conditions and time of arrival at the roost with a linear crossed mixed

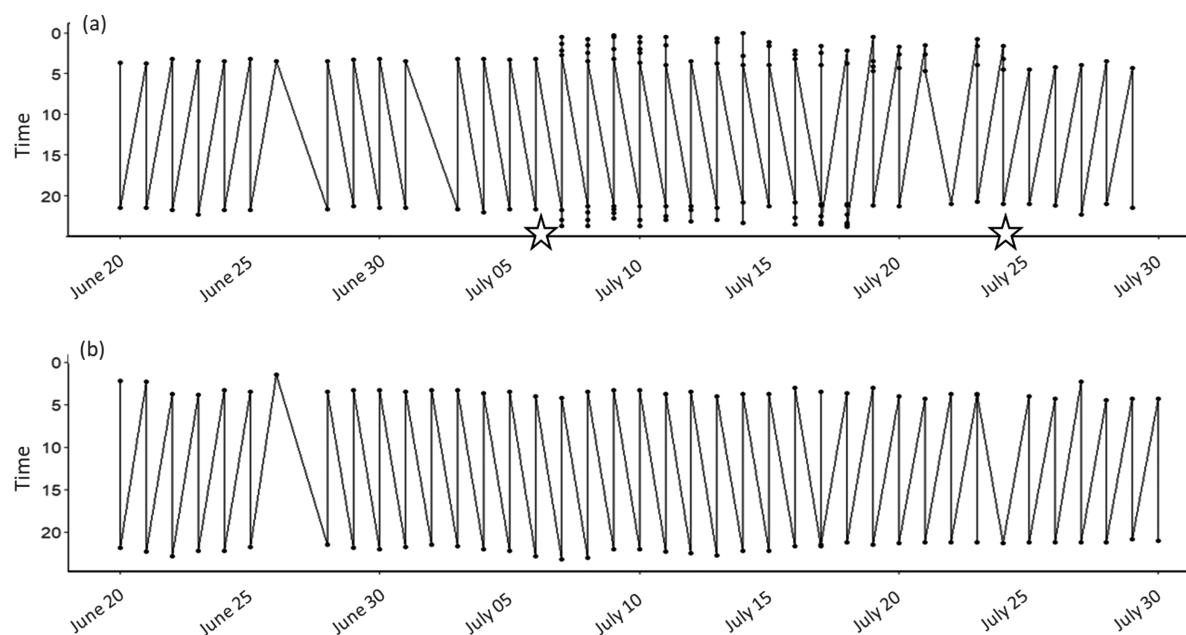


Fig. 2. Detections of PIT-tagged female bats at the Domaine Joly maternity roost (Québec, Canada) during the reproductive season in 2019. Each point represents an exit or a return to the roost. Examples of (a) a reproductive female, with an activity pattern changing from one to several visits to the roost per night after inferred parturition—the parturition and decreased parental care dates, represented by stars, are estimated to 7 July and 24 July, respectively—and (b) a nonreproductive female with an absence of activity pattern changes.

model (LMM). We first built a set of models to find the best global model. All models included the following fixed factors: a temperature metric, a rain metric, spring mean wind speed, and arrival date. Based on personal knowledge and literature on optimal temperature ranges minimizing energy expenses for reproductive bats, we tested the following metrics for temperature: mean temperature, mean temperature at night, number of hours in the optimal temperature range of 15 to 32 °C, number of hours in the optimal temperature range of 21 to 32 °C, number of days with temperature between 21 and 32 °C (Henry 2001; see also Fontaine et al. 2021 for interval definitions), and number of days with temperature between 15 and 32 °C at night (Schorr and Siemers 2021). We tested the following metrics for rainfall: total rainfall and number of days with rainfall between 2.5 and 25.0 mm (Schorr and Siemers 2021). We also tested arrival date with varying intercepts based on visual inspection of the data. From this set of models, we selected the best global model based on the Akaike's information criterion corrected for small sample size (AICc). We then used the “dredge” function of R package MuMIn (Barton 2019) on our best global model to generate a subset of models with combinations of fixed-effect terms of the best global model. We selected the most parsimonious model of this subset based on AICc and used it as a final model to interpret the effect of the different factors. We used “lme4” package in R (Bates et al. 2015) for all analyses and used confidence intervals to verify the significance of variables, i.e. if the 95% confidence intervals around the slope did not include zero.

Results

Inference of reproductive status and parturition date

Using PIT tag detection patterns, we identified the reproductive status for 90% to 97% and 63% of PIT-tagged females that were detected during both the gestation and the lactation periods at the DJ roost ($n_{\text{tot}} = 20$ to 27) and the SNP roost ($n_{\text{tot}} = 60$; Table 1), respectively. We assessed the remaining females as of unknown reproductive status (3% to 10% of females at the DJ and 36% of females at the SNP roost) because there were too many days without detections (50% of females with unknown reproductive status at DJ and 36% at SNP), too many missing detections within a day (30% of females with unknown reproductive status at the DJ and 18% at the SNP), or absence of clear patterns (e.g. a mix of

1 and several exits and returns during the whole season; 20% of females with unknown reproductive status at the DJ and 46% at the SNP). During 5 years of monitoring at the DJ, we were able to assess reproductive status of a bat both physically and from PIT tag detections 39 times. From this subset, inferred reproductive status matched reproductive status assessed from the physical inspection 92% of the time. The 8% remaining were assessed as unknown based on activity patterns because the number of days of PIT tag detections was too low to reliably assess reproductive status. Annually, we inferred parturition date for 61% to 95% of reproductive females at the DJ roost and 75% of reproductive females at the SNP roost.

Phenology and determinants of parturition date

Based on PIT tag detection inferences from 2017 to 2021, females arrived at the DJ maternity roost from 30 April to 15 June, and gave birth from 18 June to 13 July. A decrease in parental care occurred from 6 July to 3 August, with a mean of 21 days between the parturition and decreased parental care. Females departed the roost from 8 July to 15 September (Fig. 3). The proportion of individuals assessed as nonreproductive at the DJ roost varied from 4% to 15% between 2017 and 2021.

Our global model explaining parturition date included arrival date, spring mean temperature at night, number of days with rainfall between 2.5 and 25 mm during spring, and spring mean wind speed as fixed factors—and individual identity and year as random factors (Supplementary Data SD1). Two final models were within 2 AICc of the model with the lowest AICc and cumulated a weight of 0.74 (Supplementary Data SD2). The most parsimonious model included arrival date, spring mean temperature at night, and spring mean wind speed at night as fixed factors—and individual identity and year as random factors. There was a significant effect of arrival date and spring mean temperature at night on parturition date, but not of spring mean wind speed (Fig. 4). Based on the most parsimonious model (conditional r -square = 0.94), a female arriving at the roost 10 days later than the others gave birth approximately 3 days later (coef. value = 0.20, 95% CI 0.09 to 0.28; Fig. 4a). An increase of 1 °C in spring mean temperature at night was associated with females giving birth 5 days sooner (coef. value = -5.59, 95% CI -9.47 to -1.72; Fig. 4b). Parturition at the DJ roost occurred over 7 to 10 days each year but there was a 23-day range over all years combined, with much later parturition date in 2019 and earlier parturition date in 2021. Although a few individuals seemed to consistently give

Table 1. Percentage of PIT-tagged bats with inferred reproductive status, and parturition date at the Domaine Joly and Salmonier Nature Park roosts in Eastern Canada from 2017 to 2021.

Site	Year	Number of PIT-tagged females before parturition	Return rate (number of females detected at least once after being tagged)	Number of females detected during the gestation and lactation periods	Percentage of females with inferred reproductive status (reproductive/nonreproductive)	Percentage of reproductive females with inferred parturition date
Domaine Joly	2017	36	33	27	97% (93%/4%)	61%
	2018	46	31	28	97% (79%/8%)	77%
	2019	53	29	27	95% (80%/15%)	82%
	2020	53	22	21	95% (90%/5%)	95%
	2021	53	21	20	90% (75%/15%)	87%
Salmonier Nature Park	2017	1,425	N/A	60/879 ^a	63% (59%/4%)	75%

^a879 out of 1,425 bats have been detected at SNP roost in 2017, but a subsample of 60 female little brown bats was randomly selected and analyzed from those with several detections during the period of gestation and lactation (i.e. from at least June to July).

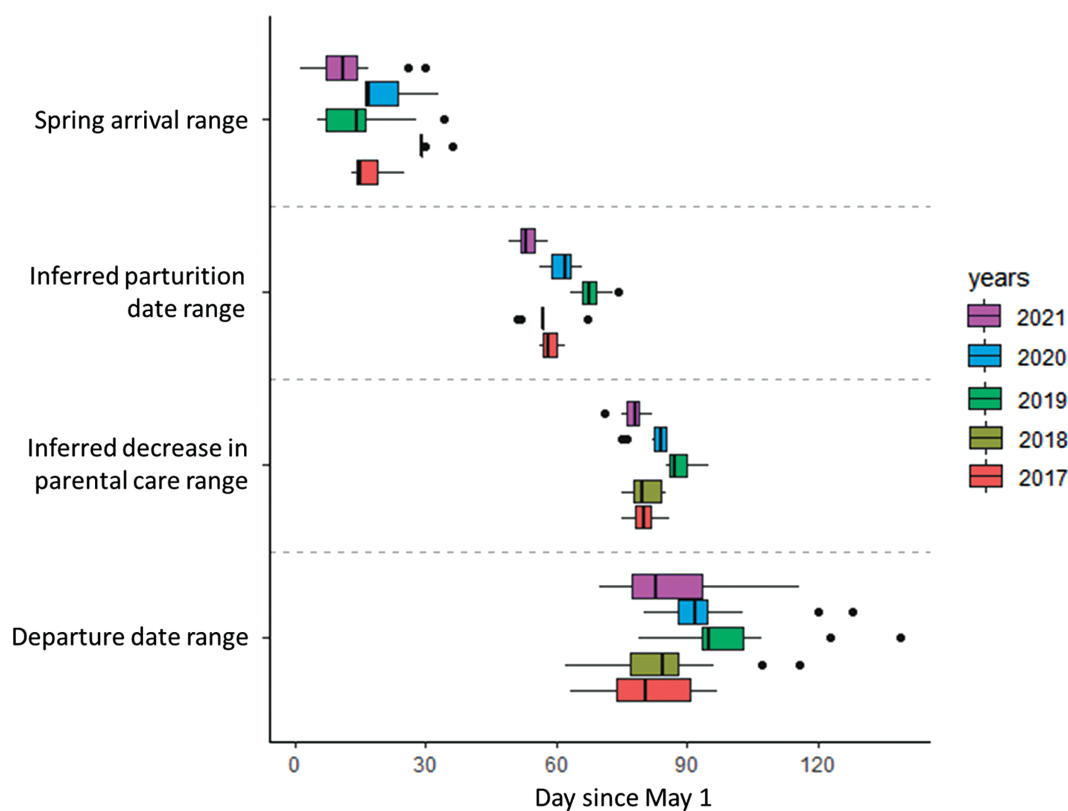


Fig. 3. Phenology of reproduction inferred from PIT tag detections for female little brown bats (*Myotis lucifugus*) and northern long-eared bats (*M. septentrionalis*) at the Domaine Joly maternity roost in Québec, Canada from 2017 to 2021. First arrival date in 2020 assumed to be prior to the date of installation (delayed due to COVID-19).

birth sooner (e.g. 859C, 6F0F, and 094E) or later (e.g. 5939, ECEE, and DAB2), most individuals displayed little consistency among years. No distinctions were detected between little brown and northern long-eared females (Fig. 5).

Discussion

Using an activity pattern change known to be typical for several species, we confirmed for the first time the potential of PIT tag systems to infer individual reproductive status and parturition date. Those reproductive inferences contribute to a better understanding of populations while being easier for researchers and less stressful for bats than other current methods such as telemetry, visual counts, or in-hand capture. Showing one of the multiple uses of these inferences, our data highlight the influence of arrival date and spring temperature on parturition date of little brown and northern long-eared bats in a northern temperate region.

Assessment of reproductive status and parturition date

Several studies have used long-term periodic captures or inspections of the roost to assess reproductive population trends (Zahn 1999; Frick et al. 2010; Francé et al. 2012; Linton and Macdonald 2018) but few identified individual parturition, lactation, and changes in parental care (e.g. Henry 2001; Frick et al. 2010). We were able to observe nightly activity patterns, and from these data, infer reproductive status and parturition date of reproductive females for most individuals at 2 roosts of *Myotis* spp. in Eastern

Canada. When detected throughout the whole season, only a few females at the DJ roost could not be assessed as reproductive or nonreproductive, mostly because of missing detections. At the SNP roost, the rate of females with an inferred reproductive status was lower, mainly because of unclear patterns. The different technologies used at the 2 sites may explain these differences, with 1 technology possibly more efficient to detect PIT-tagged bats than the other. The difference in assessment success may also be explained by roost structures. In systems where there are multiple bat boxes, individuals may move more often among the boxes and roost close to the antennas, making exits and returns more difficult to differentiate (Fontaine A, McGill University, Montreal, Quebec, Canada, personal communication, 15 September 2022). Bat boxes also have relatively large entrances allowing several bats to exit at the same time while antennas can record only 1 PIT tag at a time.

Based on our results and preliminary tests with 3 other roosts equipped with PIT tag systems in Quebec, effective inferences of reproductive parameters require individuals with high fidelity to a roost that can be effectively monitored using reliable PIT tag readers. We also advise to install antennas at the entrances of alternative day-roosts known in the vicinity in case bats temporarily switch roost due to capture disturbances. Although several limitations remain—such as PIT tag loss, antenna interferences with metallic structures, power supply issues, or inability of antennas to read several tags passing through at the same time—PIT tags have allowed innovative investigations into numerous ecological fields (Gibbons and Andrews 2004; Schorr and Siemers 2021). Recapture of bats using antennas that continuously read PIT tags can be used to get reliable inferences of multiple

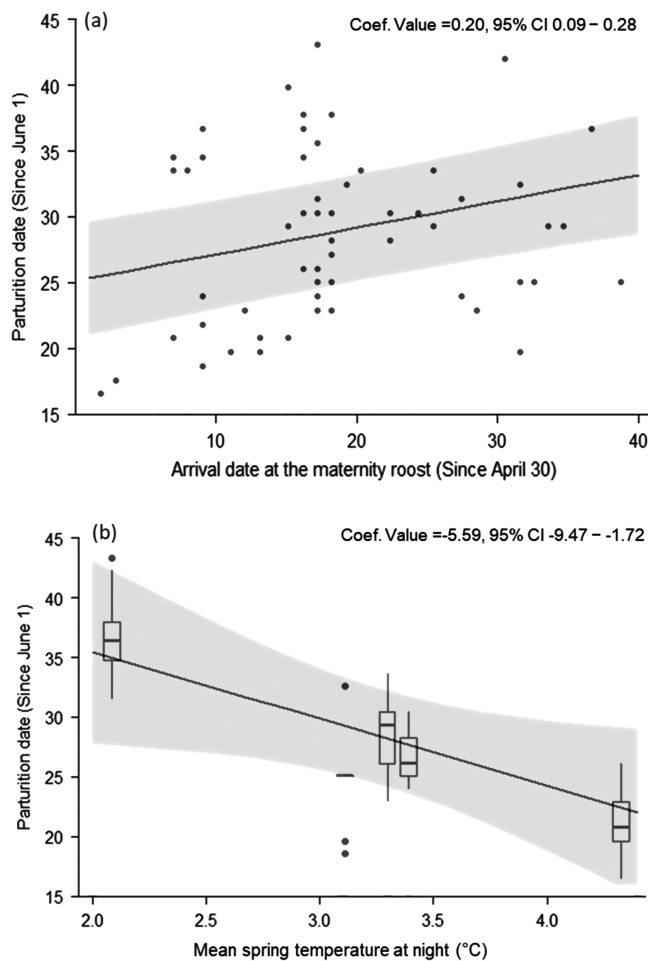


Fig. 4. Effect of (a) arrival date at the maternity roost and (b) spring mean temperature at night on parturition date of female little brown bats (*Myotis lucifugus*) and northern long-eared bats (*M. septentrionalis*) at the Domaine Joly's roost from 2017 to 2021 (Québec, Canada). The predictions are based on a linear crossed mixed model including the arrival date, spring mean temperature, and mean wind speed, with individual identity and year as random factors. Shaded areas indicate 95% confidence intervals and points show raw data of parturition date.

parameters related to phenology, reproduction, abundance, and survival (O'Shea et al. 2010; Schorr and Siemers 2021). We demonstrated here, using 2 different systems, that PIT tag detections also allow successful inferences of reproductive status and parturition date for the majority of adult females. These inferences were also validated by our in-hand captures at the DJ roost, suggesting that changes in nightly detection patterns using PIT tag records are the result of changes in reproductive state. We see great potential in this method, which could be further improved by automatized methods to clean the data or to enhance inference with imperfect PIT tag detections (Sunga 2022).

Phenology and determinants of parturition date

After emergence from hibernation in the spring, females must initiate gestation in late spring to complete reproduction before the end of summer, matching the cost of maintenance and reproduction with available resources (see Fleming and Eby 2003 for timing of migration; Jonasson and Guglielmo 2016). Female little brown bats may travel >600 km during seasonal movements from winter to summer grounds, with median distance = 463 km

in central Canada, and >400 km in Indiana and north-central Kentucky, suggesting high energetic demands (Humphrey 1971; Norquay et al. 2013). The positive relationship between individual arrival date and parturition date in our study suggests that early arrival allows for a longer breeding season that likely translates into fitness advantages for females and their pups (Frick et al. 2010). Differences in arrival and parturition dates may also be explained by interannual variations in female body condition in early spring, or intersite variations such as local conditions at hibernating and summer roosting sites, or migration distance from the hibernaculum to the maternity roost (Pettit and O'Keefe 2017; Linton and Macdonald 2018; Sommers et al. 2019). Investigating these factors would require coupling PIT tag detections with other methods such as in-hand recapture.

Determining the extent to which key life history traits, such as reproductive status and parturition date, are influenced by environmental conditions and climatic variation is crucial for predicting the consequences of climate change on long-term population dynamics and stability. During "late" springs, with prolonged suboptimal conditions, breeding females may be constrained to a trade-off between the short-term energetic benefits of torpor and the negative survival consequences of delayed parturition. Gestation length in a colony of pipistrelle bats in Scotland varied by 10 days in 2 consecutive years, with gestation being extended by cold and wet weather during the early stage (Racey and Swift 1981). Similarly, early births followed warm springs in a population of horseshoe bats (*Rhinolophus ferrumequinum*) in England, where the mean birth date was 18 days earlier following a rise of 2 °C in spring temperature (Ransome and McOwat 1994).

We observed that spring weather conditions influenced breeding phenology, with earlier parturition in warmer than in colder springs. Spring 2019 was colder than any other year observed in our study, while 2021 was the warmest. Accordingly, the latest parturition dates we found were in 2019 and the earliest were in 2021. Rain or wind did not significantly influence parturition date but more extreme conditions than those recorded in our study may influence reproduction (e.g. Grindal et al. 1992; Adams 2010). Local factors such as topography or roost type may also influence the importance of weather conditions (Willis and Brigham 2005; Lausen and Barclay 2006). According to our inferences, parturition at the DJ roost occurred over 7 to 10 days each year but there was a 23-day range over all years combined. Little Brown Bat pups were born within 15 to 30 days not too far from our study site in Québec, within 42 days in Indiana, and within 23 days in Massachusetts (Horn 1998; Henry 2001; Krochmal and Sparks 2007). Several factors such as species, colony size, internal roosting conditions, food availability, or climate-related constraints may influence birth synchrony (Hoying and Kunz 1998; Arlettaz et al. 2001; Krochmal and Sparks 2007; McKinnon et al. 2012).

A presumed decrease in parental care was inferred for many females at the DJ roost. This abrupt change in activity patterns (i.e. transition from a multi-emergence sequence back to a single sequence) could be explained by the fact that pups become volant, by a change in feeding frequency, or by the loss of a pup. The mean of 21 days between parturition and decreased parental care found in our study may be considered short for these species at this latitude, as Henry et al. (2002) found in Québec that Little Brown Bat young flew outside of the roost after 20 to 35 days. Therefore, more work is required to explain this change of activity patterns. Finally, we found interannual variation to be mainly driven by temperature and seem, in our study, more important than interindividual variation. Interindividual variation detected

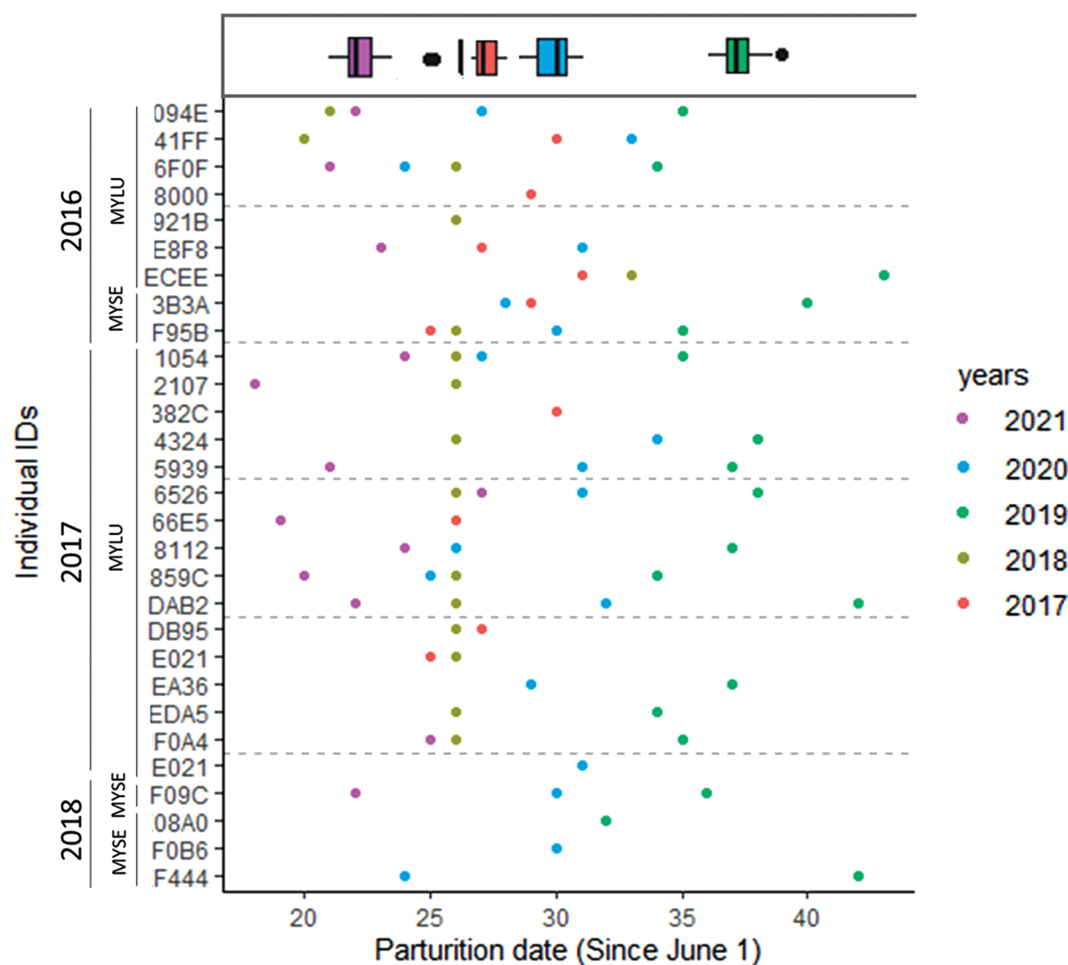


Fig. 5. Estimated parturition date of PIT-tagged female little brown bats (*Myotis lucifugus*) and northern long-eared bats (*M. septentrionalis*) at the Domaine Joly roost (Quebec Canada) from 2017 to 2021. Individuals on the y-axis ordered by ID, species, and year of marking with the mean annual parturition range at the top. Due to a malfunction of the antenna from 24 June to 28 June 2018, parturition dates that occurred during this period were all approximated as the mean of these 5 missing days.

in our data set may nonetheless be related to age or body condition (Linton and Macdonald 2018), which were not included in our analysis because they were not available for all individuals. Although we found no distinctions in parturition dates between little brown and northern long-eared bats, a bigger sample size is required to confirm that interindividual variations are not related to species (Krochmal and Sparks 2007).

PIT tag recapture methods can detect changes in individual activity pattern, and consequently, be used to infer individual reproductive status and parturition date of female bats. Although few studies evaluated the impact of PIT tags on bat health and behavior, this mark-capture-recapture method is likely valuable for reducing the behavioral response bats can have to capture and their associated biases (Ellison et al. 2007; Rigby et al. 2012; Briitzke et al. 2014; Van Harten et al. 2019). Tag loss, as is the case for all tagging methods, may vary depending on the species and impact our ability to estimate or infer parameters. As more studies are required to evaluate the full potential and risks of using PIT tags to study bats, we advise that they are only used where there is a reasonable chance of long-term monitoring (e.g. species using permanent structures as roosts and that have high fidelity to their maternity sites). As bat populations face multiple growing threats, finding ways to assess demographic variation of bat populations—as well as

its constraints—with minimal disturbance will help biologists in their conservation efforts. Our method can do so, as well as provide a detailed phenology of reproduction, which is useful to improve management actions and identify optimal timing for bat surveys. This method may also be useful in several other areas, such as in behavioral studies, and with other taxa that use hard-to-access roosts or dens.

Supplementary data

Supplementary data are available at *Journal of Mammalogy* online.

Supplementary Data SD1.—Mixed-effect model selection to determine arrival date and spring weather variables used in the global model of parturition date ($n = 80$). Individual identification and year were included as random factors in all models. Models ranked by their corresponding Akaike's information criterion corrected for small sample. Best model is bolded. If more than 1 model had $\Delta AICc < 2$, the most parsimonious model was considered best.

Supplementary Data SD2.—Top 5 of mixed-effect models of parturition date on arrival date and spring weather conditions used for model selection ($n = 80$). Individual identification and year were included as random factors in all models. Models are ranked by their corresponding Akaike's information criterion

corrected for small sample size (AICc). ΔAICc is the difference between AICc for the current model and the minimum of AICc among all the models. ω_i = Akaike weights. Best model is bolded. If more than 1 model had $\Delta\text{AICc} < 2$, the most parsimonious model was considered best.

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Author contributions

All authors contributed to the review, and the editing of this manuscript. KHE and AS contributed the supervision and funding acquisition. AS also contributed to the investigation and statistical conceptualization. VS and HGB contributed to the investigation and data curation in Quebec and Newfoundland, respectively. AF contributed to all the steps of this manuscript.

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Conflict of interest

None declared.

Data availability

Data analyzed during this study are available in the Open Science Framework repository [doi: 10.17605/OSF.IO/XVJ8E]. Full data sets available on request only.

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